



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	HW 1	Marks 50 Weight 2.5%	Due: As announced
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Important Notes

Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

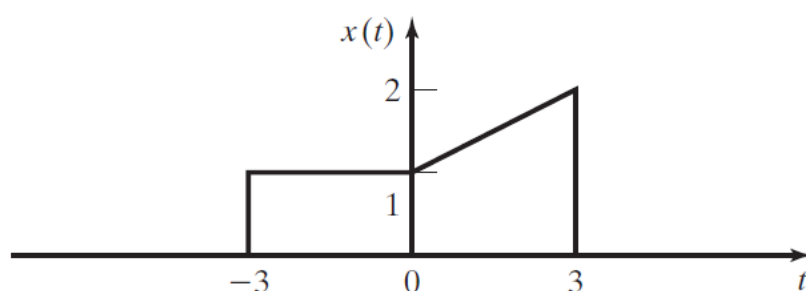
Late Submission Policy

As announced in course syllabus.

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
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Problem 1 [10 pts]

Consider the signal shown below



Plot the following

- (i) $x(-t/3)$
(iii) $x(3 + t)$

- (ii) $x(-t)$
(iv) $x(2 - t)$

Problem 2 [10 pts]

For each of the signals given, determine mathematically whether the signal is even, odd, or neither. Sketch the waveforms to verify your results.

- (a) $x(t) = -4t$
(b) $x(t) = e^{-|t|}$
(c) $x(t) = 5 \cos 3t$
(d) $x(t) = \sin(3t - \frac{\pi}{2})$
(e) $x(t) = u(t)$

Problem 3 [10 pts]

Give proofs of the following statements:

- (a) The sum of two even functions is even.
(b) The sum of two odd functions is odd.
(c) The sum of an even function and an odd function is neither even nor odd.

Problem 4 [10 pts]

Prove mathematically that the signals given below are periodic. Also find each signal's fundamental period.

- (a) $x(t) = 7 \sin 3t$
(b) $x(t) = \sin(8t + 30^\circ)$
(c) $x(t) = e^{j2t}$
(d) $x(t) = \cos t + \sin 2t$
(e) $x(t) = e^{j(5t+\pi)}$
(f) $x(t) = e^{-j10t} + e^{j15t}$

Problem 5 [10 pts]

Evaluate the following integrals

(i) $\int_{-\infty}^{\infty} \cos(2t)\delta(t)dt$

(ii) $\int_{-\infty}^{\infty} \sin(2t)\delta(t - \pi/4)dt$

(iii) $\int_{-\infty}^{\infty} \cos[2(t - \pi/4)]\delta(t - \pi/4)dt$

(iv) $\int_{-\infty}^{\infty} \sin[(t - 1)]\delta(t - 2)dt$

(v) $\int_{-\infty}^{\infty} \sin[(t - 1)]\delta(2t - 4)dt$